Revision Report: GRSL-01001-2024.R1
Title: Escape ratio contributes more than fluorescence yield to SIF-GPP relationship over crops and rainforest

Dear editors and reviewers:

We thank the editor and two reviewers for the positive comments and helpful suggestions that markedly helped to improve the manuscript. The comments from all reviewers have been carefully considered and we made our best efforts to address the comments. The point-by-point responses and revisions are provided as below. The main changes are mainly as follows.

1. We have updated the quality control data and added supplementary figures on explaining the peak growing season.

2. We have added descriptions of product details, uncertainties, and limitations.

3. We have carefully revised the language and presentation. Hope the further polished paper in terms of both the writing and organization will make it more readable.

Thank you for your precious time for reviewing our manuscript. We hope that our revised manuscript meets your expectations and look forward to hearing your decision on the publication of our manuscript.

Thank you for your attention.

Sincerely,

Wenhui Yan, Yelu Zeng, Xinhong Zhang, Weike Zhao, Yongyuan Gao, Yachang He and Dalei Hao

Comments by Editor

Comments to the Author:

- Many figures are of too low quality and should be improved in particular regarding readability of text.

Authors' reply:

Thank you for this constructive comment. We have tried our best to improve the readability of the text and the resolution of figures (900 dpi), hope these are clearer.

- It is recommended to use vector graphics for graphs and text in figures. If raster images are used, they should be of high quality. In particular visible compression artifacts should be avoided.

We have generated the images both in jpg at 900 dpi and pdf two formats to avoid visible compression artifacts.

- Please also correct the general layout of the paper, e.g. text should always be justified and not left aligned etc.

We have checked the general layout of paper and actually found Fig. 2 is left-aligned, which has been justified as required.

Associate Editor Comments:

Two reviewers provided very insightful comments. Please revise.

We thank the editor for the helpful suggestions. The detailed responses and revisions are provided in below point-by-point responses. Thank you for your precious time for reviewing our manuscript!

Reviewer #1: Comments

The manuscript by Yan et al., investigated the relationship between solar-induced chlorophyll fluorescence (SIF) and gross primary productivity (GPP), focusing on the influence of physiological and structural factors at the canopy scale. They evaluated various vegetation indices, SIF components, light use efficiency (LUE), and GPP in crops and rainforests using satellite remote sensing data. Their study revealed that the peak GPP and SIF in the United States Corn Belt were higher than in the Amazon rainforest, primarily due to higher photon escape ratios ($fesc$) and LUE in crops. They found that $fesc$, related to canopy structure, had a stronger correlation with LUE compared to fluorescence yield (ΦF), highlighting its importance in capturing LUE and GPP variations across biomes. The contribution of structural and physiological components to photosynthesis or SIF-GPP relationships is an interesting topic and the manuscript is easy to read, but there are several issues that need further attentions. Therefore, I suggest a major revision.

Authors' reply:

We thank the reviewer for the constructive comments and insightful suggestions to make the description in this paper to be more complete. We have carefully considered all of your detailed comments in the annotated version and have made revisions to address them to the best of our ability. We hope that our revisions have sufficiently addressed your concerns and improved the clarity and quality of the manuscript.

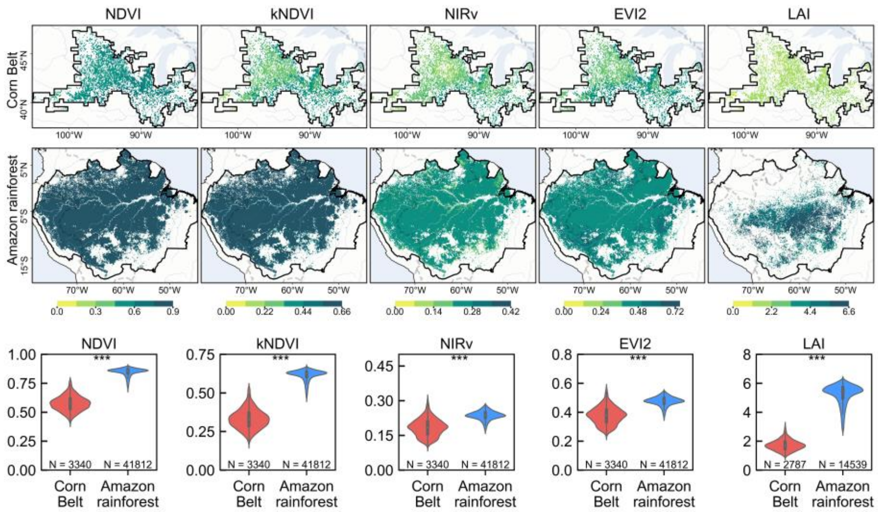
1. I have some doubts about the single month selected by the author. First, the growing seasons in the northern and southern hemispheres are different. August may be the end of the dry season for the Amazon forests, and the real growing season has not yet begun. So is it meaningful to compare the SIF values of the two regions? Secondly, August may not even be the peak of the US crop belt (maybe July?). It is recommended to select the peak of the growing season of the two regions for comparison.

Authors' reply:

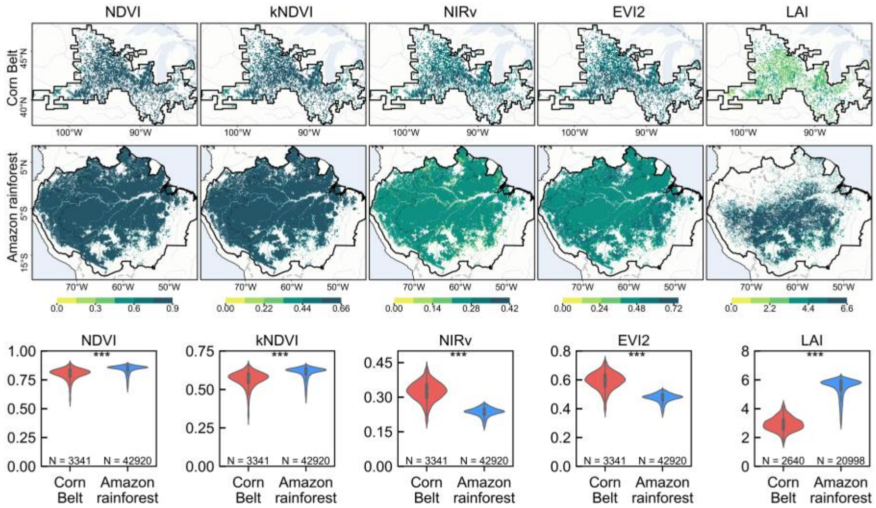
Thank you for bringing up this issue. We agree that July and August are only the peak growing season for the US Corn Belt, although not for the Amazon rainforest. We also agreed with the reviewer that the term "growing season" may be confusing especially in the northern and southern hemispheres and we appreciate the opportunity to clarify.

In the revised manuscript, we have specified that we are referring to the "peak growing season of crops in the US Corn Belt " in August. Besides, the peak growing season in the US Corn Belt is actually in July to August (USDA Crop Calendar: https://ipad.fas.usda.gov/ogamaps/cropcalendar.aspx), which were also showed in the following Supplementary Fig. 2 and Supplementary Fig. 3. The reason we have done this is to enable comparisons of different regions globally at the same time period.

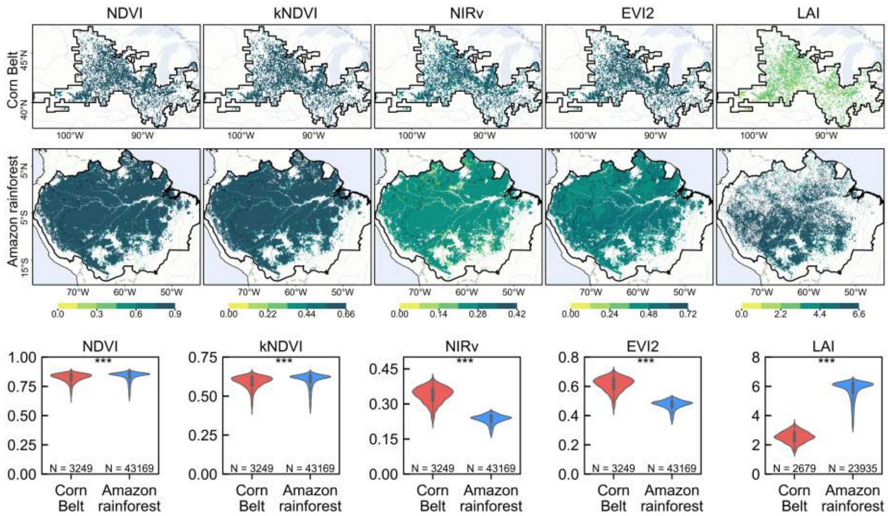
For these reasons, we chose August rather than other months. We hope this clarification will make our findings clearer and more accurate. We have also provided supplementary figures for the VI, SIF, and GPP in June, July, August and September as in below to show when this pattern developed and lasted.



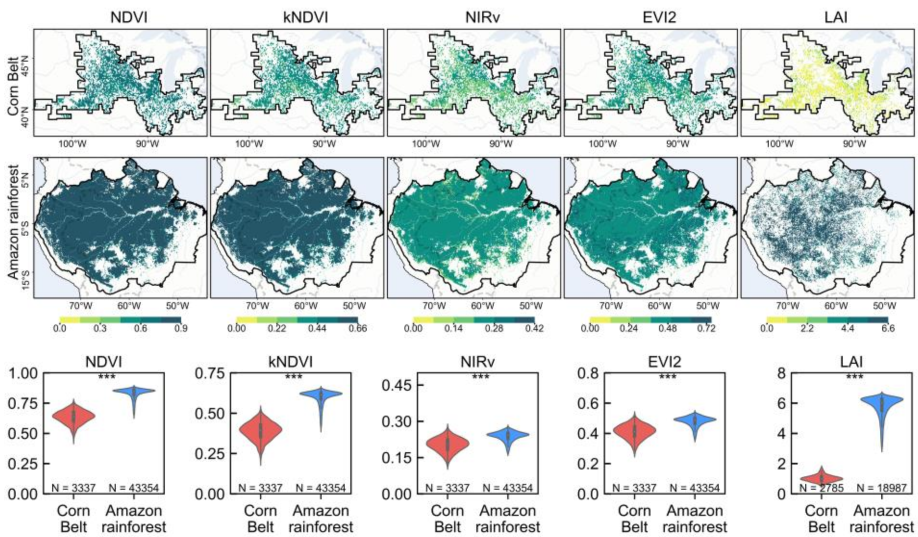
Supplementary Fig. 1. Comparison of NDVI, kNDVI, NIRv, EVI2, and LAI across the US Corn Belt and the Amazon rainforest in June for 2018-2020. The violin plots exhibit the quartiles and mean values of variables, showcasing their distribution patterns. Statistical significance level is denoted as: '***' for $p < 0.001$, '**' for $p < 0.01$, '*' for $p < 0.05$ and 'ns' for $p \geqslant 0.05$, 'N' represents the number of samples.



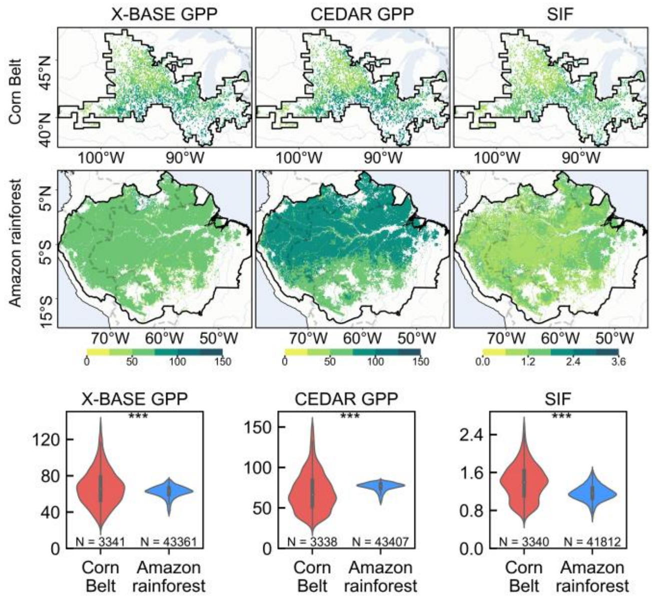
Supplementary Fig. 2. Comparison of NDVI, kNDVI, NIRv, EVI2, and LAI across the US Corn Belt and the Amazon rainforest in July for 2018-2020.



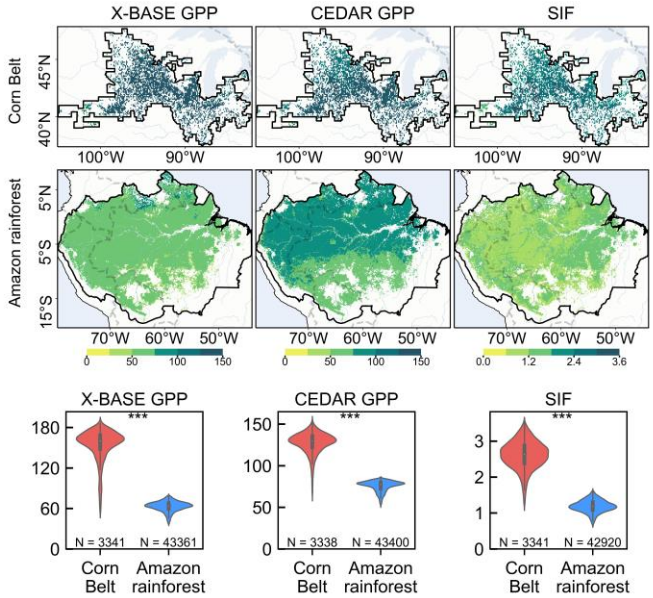
Supplementary Fig. 3 Comparison of NDVI, kNDVI, NIRv, EVI2, and LAI across the US Corn Belt and the Amazon rainforest in August for 2018-2020.



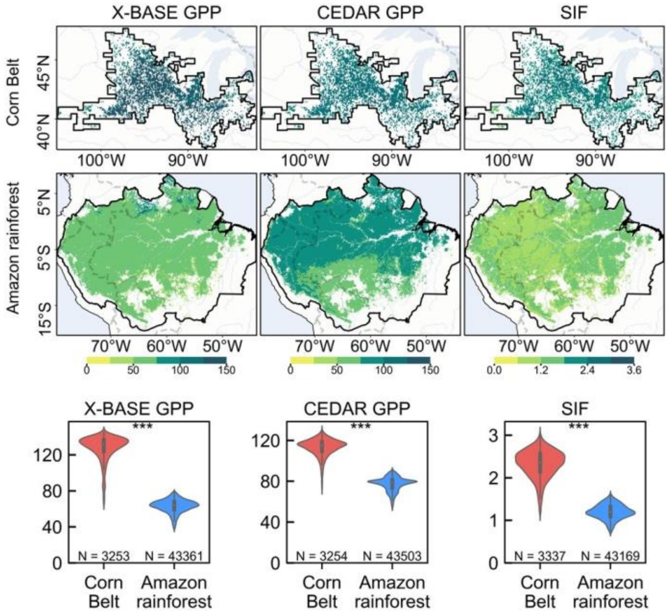
Supplementary Fig. 4. Comparison of NDVI, kNDVI, NIRv, EVI2, and LAI across the US Corn Belt and the Amazon rainforest in September for 2018-2020.



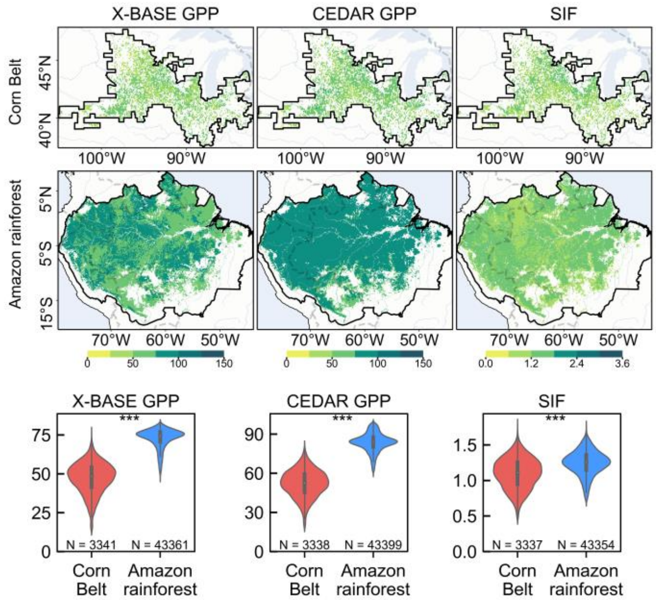
Supplementary Fig. 5. Comparison of X-BASE GPP, CEDAR GPP, and SIF across the US Corn Belt and the Amazon rainforest in June for 2018-2020.



Supplementary Fig. 6. Comparison of X-BASE GPP, CEDAR GPP, and SIF across the US Corn Belt and the Amazon rainforest in July for 2018-2020.



Supplementary Fig. 7. Comparison of X-BASE GPP, CEDAR GPP, and SIF across the US Corn Belt and the Amazon rainforest in August for 2018-2020.



Supplementary Fig. 8. Comparison of X-BASE GPP, CEDAR GPP, and SIF across the US Corn Belt and the Amazon rainforest in September for 2018-2020.

2. In spite of shorter letter, the authors still need to provide more details. For example, did the you consider quality control information for the MODIS LAI and reflectance products? This is particularly important for the Amazon, one of the study areas, where cloud and aerosol contamination of optical remote sensing data is a well-known issue. Ignoring this could affect the results. Additionally, for the TROPOMI SIF data, is it on a daily scale or instantaneous? What wavelengths were used for retrieval? The spatial resolutions of the various datasets are inconsistent; what resolution were they standardized to?

Authors' reply:

We are grateful to the valuable suggestions in refining the manuscript. In response, we considered quality control on MDODIS LAI data and have provided more details about the MODIS products and TROPOMI SIF data, such as additional information about quality control, scale and wavelengths used for retrieval.

We have used MODIS LAI and reflectance data according to the physical term definition and product standard quality, which with good Quality Assurance (QA)/ Quality Control (QC) flags. The description to support above has been revised as below in Section II.

"The MODIS MCD43A1.061 product provides the isotropic, volumetric, and geometric model parameters at Red (620–670 nm) and NIR (841-876 nm) bands to calculate reflectance at a fixed view geometry of VZA=0° (View Zenith Angle) and SZA=45° (Solar Zenith Angle) daily temporally and 500 meters spatially [15], [33]. The normalized reflectance was used to ascertain the four VIs listed in Table 1[31]."

"... We used MODIS products with good Quality Assurance (QA)/ Quality Control (QC) flags."

In the results section (Section III), we have replaced the figure mapped by quality-controlled data and changed the description about LAI (mean±s.d. has changed).

"Our research findings also demonstrate the important role of f_{esc}, indicating that high VIs and low LAI (2.53±0.38) in uniform crops but with high f_{esc} (0.47±0.04), while low VIs and high LAI (5.75±0.67) in rainforest with low f_{esc} (0.30±0.03) (Fig. 2)."

Far-red daily SIF estimates at 740 nm were retrieved from 743-758nm. We have added the description on SIF in the main text of the revised manuscript as below in Section II.

"*To ensure data quality, observations with far-red (*740nm) *daily SIF values not within −2 to 4 mW/m²/sr/nm and SIF signal uncertainty greater than 0.55 mW/m²/sr/nm were excluded.*"

As for the resolution to which various datasets were standardized, the original manuscript has illustrated this on Page 2 right, line 21-22 as below. Now the corresponding sentence in Page 2 right in revision.

"*All data were aggregated monthly temporally and 0.1° spatially, confined to August of each year from 2018 to 2020.*"

3. The authors did not directly analyze the contribution of SIF-GPP, so could the title be changed to like "fesc contributes more to the difference in photosynthesis between the two regions"?

Authors' reply:

Thank you for raising this point. This study developed from the contrast pattern that the August SIF and GPP in the US Corn Belt are higher than in the Amazon rainforest, despite the latter having a higher LAI. Then we discussed the potential relationship between SIF components (f_{esc}, Φ_F) and LUE, attempting to explain this. On the other hand, the direct observations and the final effects are reflected in the SIF and GPP. Therefore, it is necessary to include SIF and GPP in our title. That's why we've chosen to keep the original title.

Other comments

Page 1 left

Title: How about using ‘dominate’?

Authors’ reply:

Thanks for the suggestion. We truly considered using the word 'dominate' in our title at the beginning. However, we chose not to use it because physiological contributions were still not to be underestimated, especially in certain situations such as stress. Although this study did not include these aspects, using the word 'dominate' in the title may be misleading and too absolute. Instead, we prefer using “more than” to accurately explain the contributions.

Line 42-43: Existing studies indicate

Done.

Line 51: LUE should be defined first.

Authors’ reply:

Done. As in section I, Page 1 left, in the final version.

“*Studies suggest that the relationship between SIF and GPP is determined jointly by the shared term of the APAR and the SIF_{yield}-light use efficiency (LUE) relationship (SIF_{yield} =SIF/APAR) [5].*”

Page 1 right

Lines 32-34 It’s not clear how the SIF from two biomes can be compared without given time information. For example, do you mean annual SIF, growing-season mean SIF, or peak SIF? Please also check my main comment 1.

Authors’ reply:

Thanks for the insightful suggestions. We acknowledged that our description about citied paper was not clearly. We also checked your main comment #1. As in our response to a previous comment, the SIF here actually refers to the mean values during peak growing season in 2018-2020 (specifically for August). This is the time period for the reference, and also ours, for better comparisons. We have added further explanations for the cited paper in the revised manuscript to clarify the given time as below.

“... resulting in the SIF of the United States (US) Corn Belt being larger than that of the Amazon rainforest in August, despite the latter having a larger leaf area index (LAI).”

References:

Y. Zeng et al., "Structural complexity biases vegetation greenness measures," Nature Ecology & Evolution, vol. 7, no. 11, pp. 1790–1798, Nov. 2023.

Page 2 left

Study area: Please give more details about two regions and why they were selected (may also better clarify it in Introduction)

Authors’ reply:

We appreciate this valuable comment. Firstly, we highlighted different canopy structures that why the two study areas were selected in the Introduction section.

"The objectives of this study are twofold: firstly, to investigate the roles of Φ_F *and* f_{esc} *in the contrasting pattern across biomes with distinct canopy structures (i.e., the US Corn Belt and the Amazon rainforest)"*

Secondly, we described the contributions of the Amazon rainforest and the US Corn Belt to global carbon sinks and United States agriculture, respectively, in the section II.

"This biome is primarily characterized by its evergreen forest cover and stands as one of the most important land carbon sinks [1]."

"In this region, corn and soybeans are uniformly cultivated [19]."

References:

C. Beer et al., "Terrestrial Gross Carbon Dioxide Uptake: Global Distribution and Covariation with Climate," Science, vol. 329, no. 5993, pp. 834-838, Jul. 2010.

C. Boryan et al., "Monitoring US agriculture: The US Department of Agriculture, National Agricultural Statistics Service, Cropland Data Layer Program," Geocarto Int., vol. 26, no. 5, pp. 341–358, 2011.

Line 39 leverages

Done!

Page 3 left

Fig. 1 So surprised that LAI and VIs (NIRv/NDVI) totally decoupled. Could the authors further discuss why this happened considering that LAI and NDVI are highly related? Is this partly due to the MODIS data quality?

Authors' reply:

Thanks for the question. We have used quality-controlled MODIS data with good Quality Assurance (QA)/ Quality Control (QC) flags and this phenomenon cannot be attributed to data quality.

Firstly, LAI and VIs used different products, and there was no decoupling operation involved. NDVI and kNDVI is saturated when LAI is high. Besides, LAI is just one factor affecting VIs. VI can be affected by other canopy structure factors.

Secondly, generally speaking, LAI and VI are highly correlated. However, other VIs (NIRv, EVI2) in this study appeared decoupled from LAI due to the shadow effect. Relevant study has found that shadow effect caused by complex canopy structure can distort vegetation greenness. Although the Amazon rainforest has a high LAI, its VIs and SIF are lower than those of the US Corn Belt, where crops with uniform canopy structure are grown.

Page 4 left

Lines 55-57: Is this done by comparing each pixel within the regions with three August values? Can we say that spatially Fesc contributes more?

Authors' reply:

Thanks. The value of each pixel is averaged over the three months of August. Each pixel has its own Φ_F, f_{esc}, and corresponding LUE, and then the correlation analysis is performed. We can say that f_{esc} contributes more spatially.

Page 5 left

Lines 13-16: Not fair comparison since August is not the peak growing season of Amazon forests.

Authors' reply:

Thanks. As in our response to the comment #1 from another reviewer , explanation about peak growing season and selected month has discussed the fairness of this comparison.

Reviewer #2: Comments

This manuscript presents a compelling study on the relationship between solar-induced chlorophyll fluorescence (SIF) and gross primary productivity (GPP) across different biomes, providing insightful analyses and utilizing robust datasets. The authors effectively demonstrate the roles of escape ratio contributes more than fluorescence yield to SIF-GPP relationship over crops and rainforest. The use of comprehensive remote sensing products significantly contributes to our understanding of how structural and physiological canopy characteristics impact GPP estimation via SIF. The manuscript is well-structured, with clear objectives and a logical flow that enhances its readability and scientific impact. I recommend minor revisions for this manuscript.

Authors' reply:

We thank the reviewer for the positive comments on the value of our paper, as well as the helpful suggestions regarding the sensitivity analysis on the shadow effects and other factors. We have carefully considered the comments and have made corresponding revisions to improve the manuscript. Please find below the detailed responses and revisions.

1. In Figure 1, it's intriguing to note that the LAI in the Corn Belt is much lower than that in the Amazon Rainforest, yet other metrics indicate higher values for the Corn Belt compared to the Amazon. Could it be that the Corn Belt was observed during a period when crops were not fully developed or post-harvest, resulting in a lower LAI? Please provide potential reasons.

Authors' reply:

Thanks for the question. As in our response to the comment #1 raised by the reviewer # 1, the lower LAI in the Corn Belt compared to the Amazon Rainforest can be attributed to the denser canopy in the rainforest which has been demonstrated by the field measurements and satellite observations. The higher values of other VIs in the Corn Belt do not fully represent the actual vegetation greenness difference between the two regions. This difference is largely due to the shadow effect created by the complex canopy structure in the Amazon Rainforest.

References:

Y. Zeng et al., "Structural complexity biases vegetation greenness measures," Nature Ecology & Evolution, vol. 7, no. 11, pp. 1790–1798, Nov. 2023.

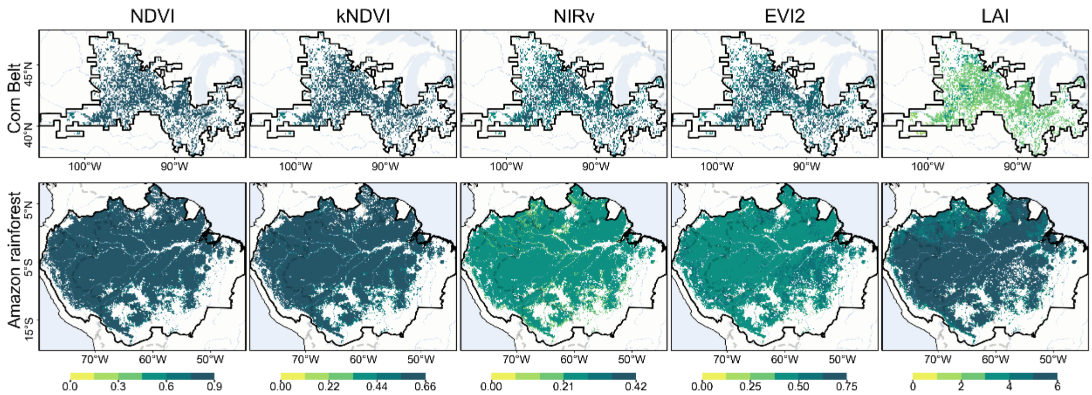
The canopy structural canopy of Amazon rainforest causes a higher ratio of photons to be intercepted and reflected within the canopy, rather than escaping, which affects the reflectivity detected by sensors. Consequently, VIs do not account for this complexity, tend to underestimate the greenness of the rainforest. In contrast, the radiative transfer models used to retrieve LAI (Leaf Area Index, is defined as the one-sided green leaf area per unit ground area in broadleaf canopies and as one-half the total needle surface area per unit ground area in coniferous canopies) takes into consideration the structural characteristics, shadows, and sun-sensor geometry, providing a more accurate measurement of vegetation greenness. Thus, the simpler and more uniform canopy structure of the Corn Belt leads to higher VI values, even though the actual leaf area is lower.

2. The spatial distribution maps in Figures 1, 2 and 3 do not clearly show the spatial differences. Consider adjusting the color bar to enhance visibility.

Authors' reply:

Thanks for the suggestion regarding the spatial distribution maps. We have adjusted the color bar to enhance the visibility of spatial differences in spatial distribution maps of Supplementary Fig. 9. Additionally, we have changed the gradient color bar more layered to better highlight the spatial differences.

We agree that the color bar in Supplementary Fig. 9 is good at taking into account the differences within and between figures, so they are used throughout the paper. We hope these adjustments will improve the clarity and effectiveness of our visualizations.



Supplementary Fig. 9. Comparison of NDVI, kNDVI, NIRv, EVI2, and LAI across the US Corn Belt and the Amazon rainforest in August for 2018-2020.

3. Since FPAR directly influences the APAR, which is a critical component in linking GPP to SIF, any uncertainties in FPAR products will propagate to those physiological and structural parameter estimations. This could affect the study's findings regarding the SIF-GPP relationship, particularly under different environmental conditions. Incorporating a discussion on these points would clarify the potential limitations of using MODIS FPAR in your study.

Authors' reply:

Thanks for the suggestion. It is indeed true that any uncertainties in FPAR products will propagate to the estimations of physiological parameters (Φ_F) and structural parameters (f_{esc}). We have now added more discussion in the Discussion section to clarify the potential limitations of using MODIS FPAR.

"Besides, product uncertainties remain, especially under different environmental conditions, and these uncertainties and their further effects should be taken into account. For example, FPAR directly influences APAR, which further affect in linking GPP to SIF."

Additionally, our research focuses on the pattern caused by structural differences between two biomes instead of short-term magnitude changes. Besides, we also utilized two GPP products to reduce the uncertainties from the data source.

References:

Yan, Kai, et al. "The Impact of Quality Control Methods on Vegetation Monitoring Using MODIS FPAR Time Series." Forests 15.3 (2024): 553.

4. Is the ERA5 PAR product used in the study a clear-sky or all-sky PAR product?

Authors' reply:

Thanks for the question. The ERA5 PAR product used in the study was an all-sky PAR product. This has been added in the revised version.

"The Photosynthetically Active Radiation (PAR) was estimated from this product in all sky conditions [12]."

5. In Figure 4, the authors have used data from the entire month of August, while the use of an 8-day FPAR product to capture crop growth may not adequately reflect the changes in crop structure.

Authors' reply:

Thanks for the question. Here we are using the average of the monthly data. Indeed, using an 8-day FPAR product to capture crop growth may not sufficiently reflect changes in crop structure. However, our study did not primarily focus on short-term changes in crop structure. Instead, we emphasize the structural differences between two biomes. Specifically, we emphasize spatial variations rather than temporal changes. Therefore, in our study, all FPAR data for the month of August were averaged to explore spatial differences in FPAR.